

IAN HAMBLIN

GRAPHICS / RENDERING ENGINEER

(WEBGL + UNITY)

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const GRAPHICS & RENDERING = [

{ skill: 'Unity SRP (URP/HDRP)', years: 5 },

{ skill: 'ShaderLab / HLSL', years: 6 },

{ skill: 'Shader Graph', years: 3 },

{ skill: 'Three.js', years: 9 },

{ skill: 'WebGL', years: 8 },

{ skill: 'Shaders (GLSL)', years: 6 },

{ skill: 'React Three Fiber', years: 4 },

{ skill: 'GPU Instancing / Batching / LOD',

years: 5 },

]

const LANGUAGES & FRAMEWORKS = [

{ skill: 'TypeScript', years: 9 },

{ skill: 'JavaScript', years: 21 },

{ skill: 'C# (Unity)', years: 7 },

{ skill: 'React', years: 5 },

{ skill: 'Node.js', years: 6 },

]

const 3D DCC & ENGINES = [

{ skill: 'Blender', years: 6 },

{ skill: '3ds Max', years: 16 },

{ skill: 'Unity', years: 7 },

{ skill: 'Unreal Engine', years: 5 },

{ skill: 'Substance Painter', years: 4 },

]

const TOOLS & DEVOPS = [

{ skill: 'Git', years: 11 },

{ skill: 'CI/CD', years: 9 },

{ skill: 'Docker', years: 6 },

{ skill: 'AWS', years: 9 },

]

EDUCATION

2004

BA Hons in Creative Visualisation, University of Teesside

1999

A Levels: Maths, Physics, Technology — St John Fisher Sixth Form

ABOUT ME

Graphics-focused engineer blending 3D art foundations with modern WebGL/Three.js and Unity rendering. Experienced shipping realtime visualisation tools and data-driven rendering for large geospatial and simulation datasets.

Comfortable across shader authoring (ShaderLab/HLSL, GLSL), Unity SRP (URP/HDRP), performance tuning, and building clean UI for technical users. Enjoys collaborating with artists and engineers to deliver visually impressive, performant, and maintainable experiences.

EMPLOYMENT HISTORY

Insight UK – Creative Developer	December 2025 – March
(Contract)	2026

- Built a high fidelity digital dashboard in Unreal Engine and Three.js, rendering complex business data as movie quality realtime visuals.
- Developed AI driven storytelling to guide viewers through complex datasets.
- Modelled and sourced the 3D assets, and authored the look and lighting of the scenes.
- Delivered to a fixed deadline for a high stakes client presentation.

3DEO – Lead Visual Engineer	2016 – 2024
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- Built a browser-based realtime 3D terrain visualizer (TypeScript + Three.js + WebGL).
- Designed colour-mapping and shader pipelines to analyse bathymetry datasets at scale.
- Reduced volumetric calc times from ~30s to 0.1s with GPU accelerated techniques.
- Implemented temporal data blending and elevation deltas using custom GLSL shaders.
- Delivered a realtime 3D flythrough showcased at COP26; integrated video and interactions.
- Worked across AWS (EC2, S3) and web stack to ship performant visual tools.

Freelance – Graphics Engineer / 3D	2011 – 2016 & 2024 –
Generalist	Present

HIGHLIGHTS

- Unity graphics pipeline: URP/HDRP setup, lighting, post-processing, render queues.
- Shader authoring: ShaderLab/HLSL (Unity) and GLSL (WebGL); surface/fragment shaders.
- Performance: tuned large scenes with LOD, GPU instancing, batching, and frustum culling.
- UX: built technical analysis tools (cross-sections, delta views) for domain experts.
- Pipelines: Blender/3ds Max tools; automated ingest and material workflows.

- Shipped 'Zippy Town 3D' end to end — an Android live wallpaper and a realtime Three.js showcase at zippy.town — to a mobile GPU budget using DRACO compressed meshes, LOD, SMAA and post processing.
- Unity C# graphics: authored shaders (ShaderLab/HLSL), configured URP/HDRP, and tuned lighting/post-processing.
- Developed Three.js based editors and realtime viewers; implemented tiling and material workflows.
- Created Python + Blender add-ons for streamlined asset export/import pipelines.
- Built VR driving simulator and multiple AR experiences (Unity + Web); integrated live sensor/geo data.
- Implemented WebRTC pipeline from browser to Blender for high-quality realtime rendering.
- Produced the photoreal hero render sequence for the M-Sport Ford Raptor T1+ Dakar site: CAD ingest, Unreal materials and lighting. The site won Awwwards Site of the Day.

Caspian Learning – Artist / Developer

2005 – 2015

- Shipped education/training titles; authored environments, characters, and animation.
- Collaborated on rigging and tooling to improve runtime performance and iteration speed.
- Introduced lightmapping workflows; built custom 3ds Max material editor for pipeline wins.